

Practical 1

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Batch: A3

import time

import random

def detect_obstacle():

Randomly simulate obstacle detection (0 = detected, 1 = not detected)

return random.choice([0, 1])

print("IR Obstacle Detection Simulation Started...")

print("Press Ctrl+C to stop.\n")

try:

while True:

n = detect_obstacle()

if n == 0:

print("OBJECT DETECTED... [RED LED ON]")

else:

print("OBJECT NOT DETECTED... [GREEN LED ON]")

time.sleep(0.5)

except KeyboardInterrupt:

print("\nSimulation stopped.")

Output

IR Obstacle Detection Simulation Started...

Press Ctrl+C to stop.

OBJECT DETECTED... [RED LED ON]

OBJECT NOT DETECTED... [GREEN LED ON]

OBJECT DETECTED... [RED LED ON]

OBJECT DETECTED... [RED LED ON]

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